

MVUE

1.0

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Chapter 1

MVUE Library

Minimum Variance Unbiased Estimator library for sensor fusion.

Minimum Variance Unbiased Estimator library for sensor fusion.

1.1 Introduction

The MVUE library provides lightweight, template-based C++ implementations of Minimum Variance Unbiased Estimators (MVUE) for combining multiple independent measurements of the same quantity.

When multiple sensors provide measurements of the same physical quantity with known measurement variances, the measurements can be combined into a single estimate with lower variance than the individual measurements.

The library provides estimators for a fixed number of measurements as well as a generic estimator supporting an arbitrary number of measurements.

1.2 Mathematical Model

Consider a quantity x measured by N independent sensors:

$$x_i = x + v_i$$

where x_i is the measurement from sensor i and v_i is zero-mean measurement noise with variance σ_i^2 .

The minimum variance unbiased estimate is obtained using inverse-variance weighting:

$$\hat{x} = \frac{\sum_{i=1}^N \frac{x_i}{\sigma_i^2}}{\sum_{i=1}^N \frac{1}{\sigma_i^2}}$$

The variance of the resulting estimate is:

$$\text{Var}(\hat{x}) = \frac{1}{\sum_{i=1}^N \frac{1}{\sigma_i^2}}$$

Therefore, measurements with lower variance receive greater weight in the resulting estimate.

The inverse-variance weighted estimator is also the Best Linear Unbiased Estimator (BLUE) when the required assumptions are satisfied.

In particular, for independent measurements with known variances, the estimator can be written as a linear combination:

$$\hat{x} = \sum_{i=1}^N w_i x_i$$

where

$$w_i = \frac{1/\sigma_i^2}{\sum_{j=1}^N 1/\sigma_j^2}.$$

The weights satisfy the unbiasedness condition:

$$\sum_{i=1}^N w_i = 1.$$

1.3 Features of Library

The library provides:

- Two-measurement MVUE estimator.
- Three-measurement MVUE estimator.
- Four-measurement MVUE estimator.
- N-measurement MVUE estimator.
- C++ template support for different numeric types.
- A lightweight interface suitable for embedded and real-time applications.

1.4 Assumptions

The estimator assumes that:

- The measurements estimate the same underlying quantity.
- The measurement errors are unbiased.
- The measurement variances are known.
- The measurement variances are positive.
- Measurements are independent when using the inverse-variance weighting formulation implemented by this library.

If measurement errors are correlated, the covariance between measurements must be considered and the simple inverse-variance formulation is no longer sufficient.

1.5 Project Structure

The library is organized as follows:

```
MVUE
+-- include
|   +-- mvue.h
|   +-- mvue_2.h
|   +-- mvue_3.h
|   +-- mvue_4.h
|   `-- mvue_N.h
|
`-- src
    +-- mvue_2.hpp
    +-- mvue_3.hpp
    +-- mvue_4.hpp
    `-- mvue_N.hpp
```

1.6 Quick Start

1.6.1 Two-Measurement Example

The following example combines two measurements with known variances:

```
#include "mvue_2.h"

estimators::mvue::MVUE_2<double> estimator;

estimator.init(0.25, 1.0);

double estimate = estimator.update(10.2, 9.8);
```

Here, the first measurement has variance 0.25 and the second measurement has variance 1.0. Therefore, the first measurement receives a larger weight.

1.6.2 Multiple Measurements

Estimators for three and four measurements are also available:

```
#include "mvue_3.h"
#include "mvue_4.h"

estimators::mvue::MVUE_3<double> estimator_3;
estimators::mvue::MVUE_4<double> estimator_4;

estimator_3.init(0.25, 1.0, 0.5);
estimator_4.init(0.25, 1.0, 0.5, 2.0);

double estimate_3 = estimator_3.update(10.2, 9.8, 10.0);
double estimate_4 = estimator_4.update(10.2, 9.8, 10.0, 10.5);
```

1.6.3 N_Measurement Estimator

For an arbitrary number of measurements, use the generic MVUE_N class:

```
#include "mvue_n.h"

constexpr std::size_t N = 5;

double variances[N] = {
    0.25,
    1.0,
    0.5,
    2.0,
    0.75
};

double measurements[N] = {
    10.2,
    9.8,
    10.0,
    10.5,
    10.1
};

estimators::mvue::MVUE_N<double, N> estimator;

estimator.init(variances);

double estimate = estimator.update(measurements);
```

1.7 API Reference

For complete documentation of the library API, see **Files/File List** in the content.

Chapter 2

mvue

Minimum Variance Unbiased Estimator

Chapter 3

Topic Index

3.1 Topics

Here is a list of all topics with brief descriptions:

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--	--------------------

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Directory Hierarchy

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Chapter 5

Namespace Index

5.1 Namespace List

Here is a list of all namespaces with brief descriptions:

estimators	
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mvue	
Contains Minimum Variance Unbiased Estimator implementations	21

Chapter 6

Class Index

6.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

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Minimum Variance Unbiased Estimator for two measurements	23
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Minimum Variance Unbiased Estimator for three measurements	27
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Minimum Variance Unbiased Estimator for four measurements	31
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Chapter 7

File Index

7.1 File List

Here is a list of all files with brief descriptions:

include/ mvue.h	
Minimum Variance Unbiased Estimator (MVUE) module	41
include/ mvue_2.h	
Two-measurement Minimum Variance Unbiased Estimator	42
include/ mvue_3.h	
Three-measurement Minimum Variance Unbiased Estimator	44
include/ mvue_4.h	
Four-measurement Minimum Variance Unbiased Estimator	46
include/ mvue_n.h	
N-measurement Minimum Variance Unbiased Estimator	49
src/ mvue_2.tpp	51
src/ mvue_3.tpp	53
src/ mvue_4.tpp	54
src/ mvue_n.tpp	56

Chapter 8

Topic Documentation

8.1 Minimum Variance Unbiased Estimators

The available implementations are:

- mvue_2 - MVUE using two measurements.
- mvue_3 - MVUE using three measurements.
- mvue_4 - MVUE using four measurements.
- mvue_N - MVUE using an arbitrary number of measurements.

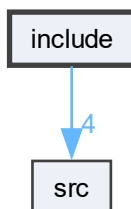
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Directory Documentation

9.1 doc Directory Reference

9.2 include Directory Reference

Directory dependency graph for include:

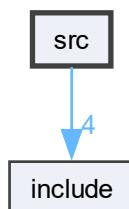


Files

- file [mvue.h](#)
Minimum Variance Unbiased Estimator (MVUE) module.
- file [mvue_2.h](#)
Two-measurement Minimum Variance Unbiased Estimator.
- file [mvue_3.h](#)
Three-measurement Minimum Variance Unbiased Estimator.
- file [mvue_4.h](#)
Four-measurement Minimum Variance Unbiased Estimator.
- file [mvue_n.h](#)
N-measurement Minimum Variance Unbiased Estimator.

9.3 src Directory Reference

Directory dependency graph for src:



Files

- file [mvue_2.hpp](#)
- file [mvue_3.hpp](#)
- file [mvue_4.hpp](#)
- file [mvue_n.hpp](#)

Chapter 10

Namespace Documentation

10.1 estimators Namespace Reference

Contains estimation and sensor-fusion algorithms.

Namespaces

- namespace [mvue](#)

10.1.1 Detailed Description

Contains estimation and sensor-fusion algorithms.

10.2 estimators::mvue Namespace Reference

Classes

- class [MVUE_2](#)
Minimum Variance Unbiased Estimator for two measurements.
- class [MVUE_3](#)
Minimum Variance Unbiased Estimator for three measurements.
- class [MVUE_4](#)
Minimum Variance Unbiased Estimator for four measurements.
- class [MVUE_N](#)
Minimum Variance Unbiased Estimator for N measurements.

10.3 mvue Namespace Reference

Contains Minimum Variance Unbiased Estimator implementations.

10.3.1 Detailed Description

Contains Minimum Variance Unbiased Estimator implementations.

Chapter 11

Class Documentation

11.1 estimators::mvue::MVUE_2< T > Class Template Reference

Minimum Variance Unbiased Estimator for two measurements.

```
#include <mvue_2.h>
```

Public Member Functions

- [MVUE_2](#) ()
Constructs a two-measurement MVUE.
- [~MVUE_2](#) ()
Destroys the MVUE estimator.
- void [init](#) (T var_1_, T var_2_)
Initializes the estimator.
- void [reset](#) ()
Resets the estimator.
- void [set_param](#) (T var_1_, T var_2_)
Sets the estimator parameters.
- T [update](#) (T x1_i, T x2_i)
Computes the minimum variance unbiased estimate.

Private Attributes

- T [var_1](#) = 0.0
Variance of the first measurement.
- T [var_2](#) = 0.0
Variance of the second measurement.

11.1.1 Detailed Description

```
template<typename T>
class estimators::mvue::MVUE_2< T >
```

Minimum Variance Unbiased Estimator for two measurements.

The [MVUE_2](#) class combines two measurements into a single estimate while minimizing the variance of the resulting estimate under the assumptions of the estimator.

Each measurement is associated with a variance. The measurement with lower variance receives a greater contribution to the final estimate.

Template Parameters

<i>T</i>	Numeric data type used for the estimator.
----------	---

Example

```
estimators::mvue::MVUE_2<double> estimator;
estimator.init(var1, var2);
double estimate = estimator.update(x1, x2);
```

Definition at line 69 of file [mvue_2.h](#).

11.1.2 Constructor & Destructor Documentation

11.1.2.1 MVUE_2()

```
template<typename T>
estimators::mvue::MVUE_2< T >::MVUE_2 ()
```

Constructs a two-measurement MVUE.

The estimator parameters are initialized to zero. Use [init\(\)](#) or [set_param\(\)](#) to configure the measurement variances before calling [update\(\)](#).

Definition at line 5 of file [mvue_2.h](#).

11.1.2.2 ~MVUE_2()

```
template<typename T>
estimators::mvue::MVUE_2< T >::~~MVUE_2 ()
```

Destroys the MVUE estimator.

Definition at line 10 of file [mvue_2.h](#).

11.1.3 Member Function Documentation

11.1.3.1 init()

```
template<typename T>
void estimators::mvue::MVUE_2< T >::init (
    T var_1_,
    T var_2_)
```

Initializes the estimator.

Parameters

in	$\var_{1_}$	Variance of the first measurement.
in	$\var_{2_}$	Variance of the second measurement.

Initializes the estimator using the variances associated with the two measurements.

Definition at line 15 of file [mvue_2.h](#).

11.1.3.2 reset()

```
template<typename T>
void estimators::mvue::MVUE_2< T >::reset ()
```

Resets the estimator.

Resets the estimator parameters to their initial values.

Definition at line 34 of file [mvue_2.h](#).

11.1.3.3 set_param()

```
template<typename T>
void estimators::mvue::MVUE_2< T >::set_param (
    T var_1_,
    T var_2_)
```

Sets the estimator parameters.

Parameters

in	$\var_{1_}$	Variance of the first measurement.
in	$\var_{2_}$	Variance of the second measurement.

Updates the variances used by the estimator when combining the two measurements.

Definition at line 21 of file [mvue_2.h](#).

11.1.3.4 update()

```
template<typename T>
T estimators::mvue::MVUE_2< T >::update (
    T x1_i,
    T x2_i)
```

Computes the minimum variance unbiased estimate.

Parameters

in	$x1_i$	First measurement.
in	$x2_i$	Second measurement.

Returns

Combined minimum variance unbiased estimate.

Combines the two input measurements using their associated variances and returns the resulting minimum variance unbiased estimate.

Definition at line 28 of file [mvue_2.h](#).

11.1.4 Member Data Documentation

11.1.4.1 var_1

```
template<typename T>
T estimators::mvue::MVUE_2< T >::var_1 = 0.0 [private]
```

Variance of the first measurement.

Definition at line 138 of file [mvue_2.h](#).

11.1.4.2 var_2

```
template<typename T>
T estimators::mvue::MVUE_2< T >::var_2 = 0.0 [private]
```

Variance of the second measurement.

Definition at line 143 of file [mvue_2.h](#).

The documentation for this class was generated from the following files:

- [include/mvue_2.h](#)
- [src/mvue_2.tpp](#)

11.2 estimators::mvue::MVUE_3< T > Class Template Reference

Minimum Variance Unbiased Estimator for three measurements.

```
#include <mvue_3.h>
```

Public Member Functions

- [MVUE_3](#) ()
Constructs a three-measurement MVUE.
- [~MVUE_3](#) ()
Destroys the MVUE estimator.
- void [init](#) (T var_1_, T var_2_, T var_3_)
Initializes the estimator.
- void [reset](#) ()
Resets the estimator.
- void [set_param](#) (T var_1_, T var_2_, T var_3_)
Sets the estimator parameters.
- T [update](#) (T x1_i, T x2_i, T x3_i)
Computes the minimum variance unbiased estimate.

Private Attributes

- T [var_1](#) = 0.0
Variance of the first measurement.
- T [var_2](#) = 0.0
Variance of the second measurement.
- T [var_3](#) = 0.0
Variance of the third measurement.

11.2.1 Detailed Description

```
template<typename T>
class estimators::mvue::MVUE_3< T >
```

Minimum Variance Unbiased Estimator for three measurements.

The [MVUE_3](#) class combines three measurements into a single estimate while minimizing the variance of the resulting estimate under the assumptions of the estimator.

Each measurement is associated with a variance. Measurements with lower variance receive greater weighting in the resulting estimate.

Template Parameters

<i>T</i>	Numeric data type used for the estimator.
----------	---

Example

```
estimators::mvue::MVUE_3<double> estimator;

estimator.init(var1, var2, var3);

double estimate = estimator.update(x1, x2, x3);
```

Definition at line 69 of file [mvue_3.h](#).

11.2.2 Constructor & Destructor Documentation

11.2.2.1 MVUE_3()

```
template<typename T>
estimators::mvue::MVUE_3< T >::MVUE_3 ()
```

Constructs a three-measurement MVUE.

The estimator parameters are initialized to zero. Use [init\(\)](#) or [set_param\(\)](#) to configure the measurement variances before calling [update\(\)](#).

Definition at line 5 of file [mvue_3.h](#).

11.2.2.2 ~MVUE_3()

```
template<typename T>
estimators::mvue::MVUE_3< T >::~~MVUE_3 ()
```

Destroys the MVUE estimator.

Definition at line 10 of file [mvue_3.h](#).

11.2.3 Member Function Documentation

11.2.3.1 init()

```
template<typename T>
void estimators::mvue::MVUE_3< T >::init (
    T var_1_,
    T var_2_,
    T var_3_)
```

Initializes the estimator.

Parameters

in	<i>var_1_</i>	Variance of the first measurement.
in	<i>var_2_</i>	Variance of the second measurement.
in	<i>var_3_</i>	Variance of the third measurement.

Initializes the estimator using the variances associated with the three measurements.

Definition at line 15 of file [mvue_3.h](#).

11.2.3.2 reset()

```
template<typename T>
void estimators::mvue::MVUE_3< T >::reset ()
```

Resets the estimator.

Resets the estimator parameters to their initial values.

Definition at line 35 of file [mvue_3.h](#).

11.2.3.3 set_param()

```
template<typename T>
void estimators::mvue::MVUE_3< T >::set_param (
    T var_1_,
    T var_2_,
    T var_3_)
```

Sets the estimator parameters.

Parameters

in	$var_{1_}$	Variance of the first measurement.
in	$var_{2_}$	Variance of the second measurement.
in	$var_{3_}$	Variance of the third measurement.

Updates the variances used by the estimator when combining the three measurements.

Definition at line 21 of file [mvue_3.h](#).

11.2.3.4 update()

```
template<typename T>
T estimators::mvue::MVUE_3< T >::update (
    T x1_i,
    T x2_i,
    T x3_i)
```

Computes the minimum variance unbiased estimate.

Parameters

in	$x1_i$	First measurement.
in	$x2_i$	Second measurement.

in	$x_3 \leftrightarrow$ _j	Third measurement.
----	-----------------------------	--------------------

Returns

Combined minimum variance unbiased estimate.

Combines the three input measurements using their associated variances and returns the resulting minimum variance unbiased estimate.

Definition at line 29 of file [mvue_3.h](#).

11.2.4 Member Data Documentation

11.2.4.1 var_1

```
template<typename T>
T estimators::mvue::MVUE_3< T >::var_1 = 0.0 [private]
```

Variance of the first measurement.

Definition at line 141 of file [mvue_3.h](#).

11.2.4.2 var_2

```
template<typename T>
T estimators::mvue::MVUE_3< T >::var_2 = 0.0 [private]
```

Variance of the second measurement.

Definition at line 146 of file [mvue_3.h](#).

11.2.4.3 var_3

```
template<typename T>
T estimators::mvue::MVUE_3< T >::var_3 = 0.0 [private]
```

Variance of the third measurement.

Definition at line 151 of file [mvue_3.h](#).

The documentation for this class was generated from the following files:

- [include/mvue_3.h](#)
- [src/mvue_3.hpp](#)

11.3 estimators::mvue::MVUE_4< T > Class Template Reference

Minimum Variance Unbiased Estimator for four measurements.

```
#include <mvue_4.h>
```

Public Member Functions

- [MVUE_4](#) ()
Constructs a four-measurement MVUE.
- [~MVUE_4](#) ()
Destroys the MVUE estimator.
- void [init](#) (T var_1_, T var_2_, T var_3_, T var_4_)
Initializes the estimator.
- void [reset](#) ()
Resets the estimator.
- void [set_param](#) (T var_1_, T var_2_, T var_3_, T var_4_)
Sets the estimator parameters.
- T [update](#) (T x1_i, T x2_i, T x3_i, T x4_i)
Computes the minimum variance unbiased estimate.

Private Attributes

- T [var_1](#) = 0.0
Variance of the first measurement.
- T [var_2](#) = 0.0
Variance of the second measurement.
- T [var_3](#) = 0.0
Variance of the third measurement.
- T [var_4](#) = 0.0
Variance of the fourth measurement.

11.3.1 Detailed Description

```
template<typename T>
class estimators::mvue::MVUE_4< T >
```

Minimum Variance Unbiased Estimator for four measurements.

The [MVUE_4](#) class combines four measurements into a single estimate while minimizing the variance of the resulting estimate under the assumptions of the estimator.

Each measurement is associated with a variance. Measurements with lower variance receive greater weighting in the resulting estimate.

Template Parameters

<i>T</i>	Numeric data type used for the estimator.
----------	---

Example

```
estimators::mvue::MVUE_4<double> estimator;

estimator.init(var1, var2, var3, var4);

double estimate = estimator.update(x1, x2, x3, x4);
```

Definition at line 69 of file [mvue_4.h](#).

11.3.2 Constructor & Destructor Documentation

11.3.2.1 MVUE_4()

```
template<typename T>
estimators::mvue::MVUE_4< T >::MVUE_4 ()
```

Constructs a four-measurement MVUE.

The estimator parameters are initialized to zero. Use [init\(\)](#) or [set_param\(\)](#) to configure the measurement variances before calling [update\(\)](#).

Definition at line 5 of file [mvue_4.h](#).

11.3.2.2 ~MVUE_4()

```
template<typename T>
estimators::mvue::MVUE_4< T >::~~MVUE_4 ()
```

Destroys the MVUE estimator.

Definition at line 10 of file [mvue_4.h](#).

11.3.3 Member Function Documentation

11.3.3.1 init()

```
template<typename T>
void estimators::mvue::MVUE_4< T >::init (
    T var_1_,
    T var_2_,
    T var_3_,
    T var_4_)
```

Initializes the estimator.

Parameters

in	<i>var_1_</i>	Variance of the first measurement.
in	<i>var_2_</i>	Variance of the second measurement.
in	<i>var_3_</i>	Variance of the third measurement.
in	<i>var_4_</i>	Variance of the fourth measurement.

Initializes the estimator using the variances associated with the four measurements.

Definition at line 15 of file [mvue_4.h](#).

11.3.3.2 reset()

```
template<typename T>
void estimators::mvue::MVUE_4< T >::reset ()
```

Resets the estimator.

Resets the estimator parameters to their initial values.

Definition at line 36 of file [mvue_4.h](#).

11.3.3.3 set_param()

```
template<typename T>
void estimators::mvue::MVUE_4< T >::set_param (
    T var_1_,
    T var_2_,
    T var_3_,
    T var_4_)
```

Sets the estimator parameters.

Parameters

in	$\var_{1_}$	Variance of the first measurement.
in	$\var_{2_}$	Variance of the second measurement.
in	$\var_{3_}$	Variance of the third measurement.
in	$\var_{4_}$	Variance of the fourth measurement.

Updates the variances used by the estimator when combining the four measurements.

Definition at line 21 of file [mvue_4.h](#).

11.3.3.4 update()

```
template<typename T>
T estimators::mvue::MVUE_4< T >::update (
    T x1_i,
    T x2_i,
    T x3_i,
    T x4_i)
```

Computes the minimum variance unbiased estimate.

Parameters

in	$x1 \leftrightarrow$ _i	First measurement.
in	$x2 \leftrightarrow$ _i	Second measurement.
in	$x3 \leftrightarrow$ _i	Third measurement.
in	$x4 \leftrightarrow$ _i	Fourth measurement.

Returns

Combined minimum variance unbiased estimate.

Combines the four input measurements using their associated variances and returns the resulting minimum variance unbiased estimate.

Definition at line 30 of file [mvue_4.h](#).

11.3.4 Member Data Documentation**11.3.4.1 var_1**

```
template<typename T>
T estimators::mvue::MVUE_4< T >::var_1 = 0.0 [private]
```

Variance of the first measurement.

Definition at line 144 of file [mvue_4.h](#).

11.3.4.2 var_2

```
template<typename T>
T estimators::mvue::MVUE_4< T >::var_2 = 0.0 [private]
```

Variance of the second measurement.

Definition at line 149 of file [mvue_4.h](#).

11.3.4.3 var_3

```
template<typename T>
T estimators::mvue::MVUE_4< T >::var_3 = 0.0 [private]
```

Variance of the third measurement.

Definition at line 154 of file [mvue_4.h](#).

11.3.4.4 var_4

```
template<typename T>
T estimators::mvue::MVUE_4< T >::var_4 = 0.0 [private]
```

Variance of the fourth measurement.

Definition at line 159 of file [mvue_4.h](#).

The documentation for this class was generated from the following files:

- [include/mvue_4.h](#)
- [src/mvue_4.tpp](#)

11.4 estimators::mvue::MVUE_N< T, N > Class Template Reference

Minimum Variance Unbiased Estimator for N measurements.

```
#include <mvue_n.h>
```

Public Member Functions

- [MVUE_N](#) ()
Constructs an N-measurement MVUE.
- [~MVUE_N](#) ()
Destroys the N-measurement MVUE.
- void [init](#) (T *var_)
Initializes the estimator.
- void [reset](#) ()
Resets the estimator.
- void [set_param](#) (T *var_)
Sets the estimator parameters.
- T [update](#) (const T *x_i)
Computes the minimum variance unbiased estimate.

Private Attributes

- T * [var](#)
Array containing the measurement variances.

11.4.1 Detailed Description

```
template<typename T, std::size_t N>
class estimators::mvue::MVUE_N< T, N >
```

Minimum Variance Unbiased Estimator for N measurements.

The [MVUE_N](#) class combines N measurements into a single estimate while minimizing the variance of the resulting estimate under the assumptions of the estimator.

Each measurement has an associated variance. Measurements with lower variance receive greater weighting in the resulting estimate.

The number of measurements is specified at compile time through the N template parameter.

Template Parameters

<i>T</i>	Numeric data type used for the estimator.
<i>N</i>	Number of measurements.

Example

```
constexpr std::size_t N = 5;

estimators::mvue::MVUE_N<double, N> estimator;

double variances[N] = {
    var1, var2, var3, var4, var5
};

estimator.init(variances);

double measurements[N] = {
    x1, x2, x3, x4, x5
};

double estimate = estimator.update(measurements);
```

Definition at line 88 of file [mvue_n.h](#).

11.4.2 Constructor & Destructor Documentation

11.4.2.1 MVUE_N()

```
template<typename T, std::size_t N>
estimators::mvue::MVUE_N< T, N >::MVUE_N ()
```

Constructs an N-measurement MVUE.

Constructs the estimator for N measurements. The estimator parameters should be initialized using [init\(\)](#) or [set_param\(\)](#) before calling [update\(\)](#).

Definition at line 5 of file [mvue_n.h](#).

11.4.2.2 ~MVUE_N()

```
template<typename T, std::size_t N>
estimators::mvue::MVUE_N< T, N >::~~MVUE_N ()
```

Destroys the N-measurement MVUE.

Definition at line 10 of file [mvue_n.h](#).

11.4.3 Member Function Documentation

11.4.3.1 init()

```
template<typename T, std::size_t N>
void estimators::mvue::MVUE_N< T, N >::init (
    T * var_)
```

Initializes the estimator.

Parameters

in	<i>var_↔</i>	Array containing the variance of each measurement.
	—	

The array must contain N variance values, where `var_[i]` represents the variance associated with the measurement `x_i[i]`.

Warning

The caller must ensure that `var_` points to valid storage containing at least N elements.

Definition at line 15 of file [mvue_n.h](#).

11.4.3.2 reset()

```
template<typename T, std::size_t N>
void estimators::mvue::MVUE_N< T, N >::reset ()
```

Resets the estimator.

Resets the estimator parameters to their initial state.

Definition at line 42 of file [mvue_n.h](#).

11.4.3.3 set_param()

```
template<typename T, std::size_t N>
void estimators::mvue::MVUE_N< T, N >::set_param (
    T * var_)
```

Sets the estimator parameters.

Parameters

in	$\text{var}_\leftrightarrow$	Array containing the variance of each measurement.
	—	

Updates the variances used by the estimator.

The array must contain N variance values.

Warning

The caller must ensure that `var_` points to valid storage containing at least N elements.

Definition at line 21 of file [mvue_n.h](#).

11.4.3.4 update()

```
template<typename T, std::size_t N>
T estimators::mvue::MVUE_N< T, N >::update (
    const T * x_i)
```

Computes the minimum variance unbiased estimate.

Parameters

in	$x_\leftrightarrow$	Array containing the measurements to be combined.
	$_i$	

Returns

Combined minimum variance unbiased estimate.

Combines the N input measurements using their associated variances and returns the resulting minimum variance unbiased estimate.

The input array must contain at least N measurements.

Warning

The caller must ensure that `x_i` points to valid storage containing at least N elements.

Definition at line 27 of file [mvue_n.h](#).

11.4.4 Member Data Documentation

11.4.4.1 var

```
template<typename T, std::size_t N>  
T* estimators::mvue::MVUE_N< T, N >::var [private]
```

Array containing the measurement variances.

Stores the variance associated with each of the N measurements.

Definition at line 176 of file [mvue_n.h](#).

The documentation for this class was generated from the following files:

- [include/mvue_n.h](#)
- [src/mvue_n.hpp](#)

Chapter 12

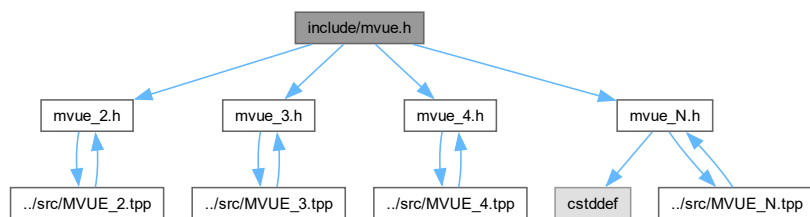
File Documentation

12.1 doc/mainpage.dox File Reference

12.2 include/mvue.h File Reference

Minimum Variance Unbiased Estimator (MVUE) module.

```
#include "mvue_2.h"  
#include "mvue_3.h"  
#include "mvue_4.h"  
#include "mvue_N.h"  
Include dependency graph for mvue.h:
```



12.2.1 Detailed Description

Minimum Variance Unbiased Estimator (MVUE) module.

This header provides the complete MVUE estimator interface by including implementations for different numbers of measurements.

Definition in file [mvue.h](#).

12.3 mvue.h

[Go to the documentation of this file.](#)

```

00001
00020
00024
00025 #ifndef ESTIMATORS_MVUE_H_BLUE_H
00026 #define ESTIMATORS_MVUE_H_BLUE_H
00027
00028 #include "mvue_2.h"
00029 #include "mvue_3.h"
00030 #include "mvue_4.h"
00031 #include "mvue_N.h"
00032
00033 #endif // ESTIMATORS_MVUE_H_BLUE_H
00034

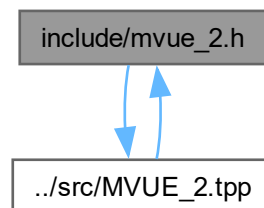
```

12.4 include/mvue_2.h File Reference

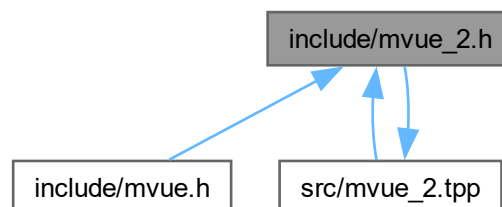
Two-measurement Minimum Variance Unbiased Estimator.

```
#include "../src/MVUE_2.tpp"
```

Include dependency graph for mvue_2.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [estimators::mvue::MVUE_2< T >](#)
Minimum Variance Unbiased Estimator for two measurements.

Namespaces

- namespace [estimators](#)
Contains estimation and sensor-fusion algorithms.
- namespace [estimators::mvue](#)
- namespace [mvue](#)
Contains Minimum Variance Unbiased Estimator implementations.

12.4.1 Detailed Description

Two-measurement Minimum Variance Unbiased Estimator.

This file defines the [estimators::mvue::MVUE_2](#) class, which implements a Minimum Variance Unbiased Estimator (MVUE) for combining two independent measurements.

The estimator combines the two input measurements according to their associated variances. Measurements with lower variance are assigned greater weight in the resulting estimate.

The class is templated to support different numeric types such as `float` and `double`.

Template Parameters

<i>T</i>	Numeric data type used for the estimator calculations.
----------	--

Author

Abhinay Kumar

Version

1.0

Date

2026-08-11

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Definition in file [mvue_2.h](#).

12.5 mvue_2.h

[Go to the documentation of this file.](#)

```

00001
00026
00027 #ifndef ESTIMATORS_MVUE_2_H_BLUE_2_H
00028 #define ESTIMATORS_MVUE_2_H_BLUE_2_H
00029
00030 namespace estimators
00031 {
00036
00037     namespace mvue
00038     {
00043
00068         template <typename T>
00069         class MVUE_2
00070         {
00071         public:
00080             MVUE_2();
00081
00085             ~MVUE_2();
00086
00097             void init(T var_1_, T var_2_);
00098
00109             void set_param(T var_1_, T var_2_);
00110
00124             T update(T x1_i, T x2_i);
00125
00132             void reset();
00133
00134         private:
00138             T var_1 = 0.0;
00139
00143             T var_2 = 0.0;
00144         };
00145
00146 #include "../src/MVUE_2.tpp"
00147
00148     } // namespace mvue
00149
00150 } // namespace estimators
00151
00152 #endif // ESTIMATORS_MVUE_2_H_BLUE_2_H

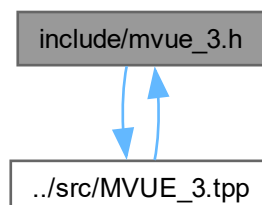
```

12.6 include/mvue_3.h File Reference

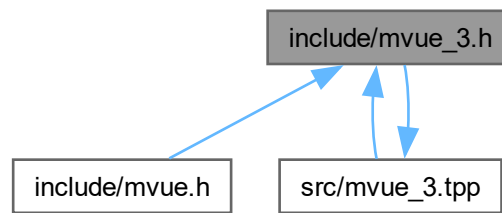
Three-measurement Minimum Variance Unbiased Estimator.

```
#include "../src/MVUE_3.tpp"
```

Include dependency graph for mvue_3.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [estimators::mvue::MVUE_3< T >](#)
Minimum Variance Unbiased Estimator for three measurements.

Namespaces

- namespace [estimators](#)
Contains estimation and sensor-fusion algorithms.
- namespace [estimators::mvue](#)
- namespace [mvue](#)
Contains Minimum Variance Unbiased Estimator implementations.

12.6.1 Detailed Description

Three-measurement Minimum Variance Unbiased Estimator.

This file defines the [estimators::mvue::MVUE_3](#) class, which implements a Minimum Variance Unbiased Estimator (MVUE) for combining three measurements.

Each measurement is associated with a variance. The estimator combines the measurements according to their variances, giving greater weight to measurements with lower variance.

The class is templated to support different numeric types such as `float` and `double`.

Template Parameters

<i>T</i>	Numeric data type used for the estimator calculations.
----------	--

Author

Abhinay Kumar

Version

1.0

Date

2026-08-11

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Definition in file [mvue_3.h](#).

12.7 mvue_3.h

[Go to the documentation of this file.](#)

```

00001
00026
00027 #ifndef ESTIMATORS_MVUE_3_H_BLUE_3_H
00028 #define ESTIMATORS_MVUE_3_H_BLUE_3_H
00029
00030 namespace estimators
00031 {
00036
00037     namespace mvue
00038     {
00043
00068         template <typename T>
00069         class MVUE_3
00070         {
00071         public:
00080             MVUE_3();
00081
00085             ~MVUE_3();
00086
00098             void init(T var_1_, T var_2_, T var_3_);
00099
00111             void set_param(T var_1_, T var_2_, T var_3_);
00112
00127             T update(T x1_i, T x2_i, T x3_i);
00128
00135             void reset();
00136
00137         private:
00141             T var_1 = 0.0;
00142
00146             T var_2 = 0.0;
00147
00151             T var_3 = 0.0;
00152         };
00153
00154 #include "../src/MVUE_3.tpp"
00155
00156     } // namespace mvue
00157 } // namespace estimators
00159
00160 #endif // ESTIMATORS_MVUE_3_H_BLUE_3_H

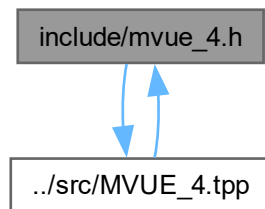
```

12.8 include/mvue_4.h File Reference

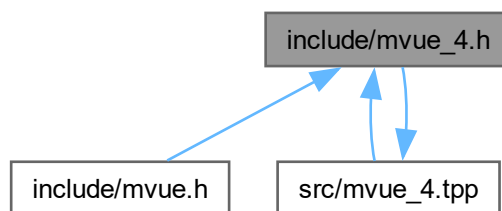
Four-measurement Minimum Variance Unbiased Estimator.

```
#include "../src/MVUE_4.hpp"
```

Include dependency graph for mvue_4.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [estimators::mvue::MVUE_4< T >](#)
Minimum Variance Unbiased Estimator for four measurements.

Namespaces

- namespace [estimators](#)
Contains estimation and sensor-fusion algorithms.
- namespace [estimators::mvue](#)
- namespace [mvue](#)
Contains Minimum Variance Unbiased Estimator implementations.

12.8.1 Detailed Description

Four-measurement Minimum Variance Unbiased Estimator.

This file defines the `estimators::mvue::MVUE_4` class, which implements a Minimum Variance Unbiased Estimator (MVUE) for combining four measurements.

Each measurement is associated with a variance. The estimator combines the measurements according to their variances, giving greater weight to measurements with lower variance.

The class is templated to support different numeric types such as `float` and `double`.

Template Parameters

<code>T</code>	Numeric data type used for the estimator calculations.
----------------	--

Author

Abhinay Kumar

Version

1.0

Date

2026-08-11

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Definition in file `mvue_4.h`.

12.9 mvue_4.h

[Go to the documentation of this file.](#)

```

00001
00026
00027 #ifndef ESTIMATORS_MVUE_4_H_BLUE_4_H
00028 #define ESTIMATORS_MVUE_4_H_BLUE_4_H
00029
00030 namespace estimators
00031 {
00036
00037     namespace mvue
00038     {
00043
00068         template <typename T>
00069         class MVUE_4
00070         {
00071         public:
00080             MVUE_4();
00081
00085             ~MVUE_4();
00086
00099             void init(T var_1_, T var_2_, T var_3_, T var_4_);
00100

```

```

00113         void set_param(T var_1_, T var_2_, T var_3_, T var_4_);
00114
00130         T update(T x1_i, T x2_i, T x3_i, T x4_i);
00131
00138         void reset();
00139
00140     private:
00144         T var_1 = 0.0;
00145
00149         T var_2 = 0.0;
00150
00154         T var_3 = 0.0;
00155
00159         T var_4 = 0.0;
00160     };
00161
00162 #include "../src/MVUE_4.tpp"
00163
00164     } // namespace mvue
00165
00166 } // namespace estimators
00167
00168 #endif // ESTIMATORS_MVUE_4_H_BLUE_4_H

```

12.10 include/mvue_n.h File Reference

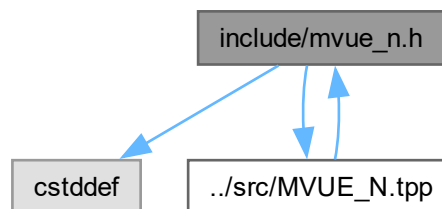
N-measurement Minimum Variance Unbiased Estimator.

```

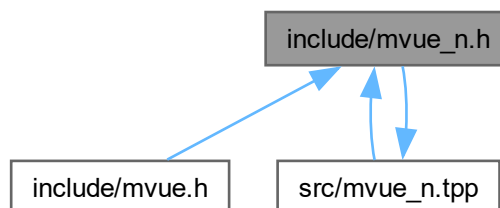
#include <cstdint>
#include "../src/MVUE_N.tpp"

```

Include dependency graph for mvue_n.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [estimators::mvue::MVUE_N< T, N >](#)
Minimum Variance Unbiased Estimator for N measurements.

Namespaces

- namespace [estimators](#)
Contains estimation and sensor-fusion algorithms.
- namespace [estimators::mvue](#)
- namespace [mvue](#)
Contains Minimum Variance Unbiased Estimator implementations.

12.10.1 Detailed Description

N-measurement Minimum Variance Unbiased Estimator.

This file defines the [estimators::mvue::MVUE_N](#) class, which implements a Minimum Variance Unbiased Estimator (MVUE) for combining an arbitrary number of measurements.

Each measurement is associated with a variance. The estimator combines the measurements according to their variances, giving greater weight to measurements with lower variance.

Unlike the fixed-size MVUE_2, MVUE_3, and MVUE_4 implementations, MVUE_N supports a compile-time configurable number of measurements.

The class is templated to support different numeric types and different numbers of measurements.

Template Parameters

<i>T</i>	Numeric data type used for the estimator calculations.
<i>N</i>	Number of measurements to be combined.

Author

Abhinay Kumar

Version

1.0

Date

2026-08-11

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Definition in file [mvue_n.h](#).

12.11 mvue_n.h

[Go to the documentation of this file.](#)

```

00001
00030
00031 #ifndef ESTIMATORS_MVUE_N_H_BLUE_4_H
00032 #define ESTIMATORS_MVUE_N_H_BLUE_4_H
00033
00034 #include <stddef>
00035
00036 namespace estimators
00037 {
00042     namespace mvue
00043     {
00044         {
00049
00087         template <typename T, std::size_t N>
00088         class MVUE_N
00089         {
00090         public:
00099             MVUE_N();
00100
00104             ~MVUE_N();
00105
00121             void init(T *var_);
00122
00138             void set_param(T *var_);
00139
00158             T update(const T *x_i);
00159
00166             void reset();
00167
00168         private:
00176             T *var;
00177         };
00178
00179 #include "../src/MVUE_N.tpp"
00180
00181     } // namespace mvue
00182
00183 } // namespace estimators
00184
00185 #endif // ESTIMATORS_MVUE_N_H_BLUE_4_H

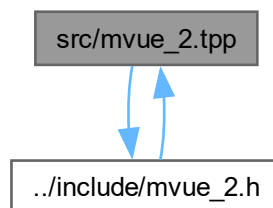
```

12.12 README.md File Reference

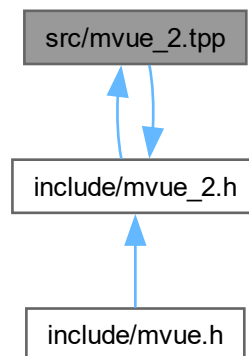
12.13 src/mvue_2.tpp File Reference

```
#include "../include/mvue_2.h"
```

Include dependency graph for mvue_2.tpp:



This graph shows which files directly or indirectly include this file:



12.14 mvue_2.hpp

[Go to the documentation of this file.](#)

```

00001 #include "../include/mvue_2.h"
00002
00003 template <typename T>
00004 estimators::mvue::MVUE_2<T>::MVUE_2()
00005 {
00006 }
00007
00008 template <typename T>
00009 estimators::mvue::MVUE_2<T>::~~MVUE_2()
00010 {
00011 }
00012
00013 template <typename T>
00014 void estimators::mvue::MVUE_2<T>::init(T var_1_, T var_2_)
00015 {
00016     set_param(var_1_, var_2_);
00017 }
00018
00019 template <typename T>
00020 void estimators::mvue::MVUE_2<T>::set_param(T var_1_, T var_2_)
00021 {
00022     var_1 = var_1_;
00023     var_2 = var_2_;
00024 }
00025
00026 template <typename T>
00027 T estimators::mvue::MVUE_2<T>::update(T x1_i, T x2_i)
00028 {
00029     return ((x1_i / var_1) + (x2_i / var_2)) / ((1.0 / var_1) + (1.0 / var_2));
00030 }
00031
00032 template <typename T>
00033 void estimators::mvue::MVUE_2<T>::reset()
00034 {
00035 }
  
```

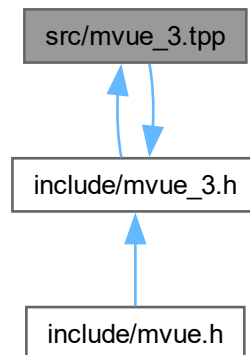
12.15 src/mvue_3.hpp File Reference

```
#include "../include/mvue_3.h"
```

Include dependency graph for mvue_3.hpp:



This graph shows which files directly or indirectly include this file:



12.16 mvue_3.hpp

[Go to the documentation of this file.](#)

```

00001 #include "../include/mvue_3.h"
00002
00003 template <typename T>
00004 estimators::mvue::MVUE_3<T>::MVUE_3()
00005 {
00006 }
00007
00008 template <typename T>
00009 estimators::mvue::MVUE_3<T>::~~MVUE_3()
00010 {
00011 }
00012
00013 template <typename T>

```

```

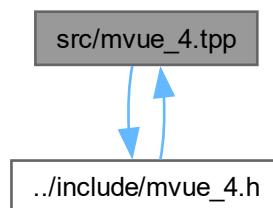
00014 void estimators::mvue::MVUE_3<T>::init(T var_1_, T var_2_, T var_3_)
00015 {
00016     set_param(var_1_, var_2_, var_3_);
00017 }
00018
00019 template <typename T>
00020 void estimators::mvue::MVUE_3<T>::set_param(T var_1_, T var_2_, T var_3_)
00021 {
00022     var_1 = var_1_;
00023     var_2 = var_2_;
00024     var_3 = var_3_;
00025 }
00026
00027 template <typename T>
00028 T estimators::mvue::MVUE_3<T>::update(T x1_i, T x2_i, T x3_i)
00029 {
00030     return ((x1_i / var_1) + (x2_i / var_2) + (x3_i / var_3)) / ((1.0 / var_1) + (1.0 / var_2) + (1.0
/ var_3));
00031 }
00032
00033 template <typename T>
00034 void estimators::mvue::MVUE_3<T>::reset()
00035 {
00036 }

```

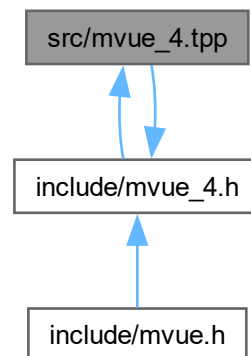
12.17 src/mvue_4.tpp File Reference

```
#include "../include/mvue_4.h"
```

Include dependency graph for mvue_4.tpp:



This graph shows which files directly or indirectly include this file:



12.18 mvue_4.tpp

[Go to the documentation of this file.](#)

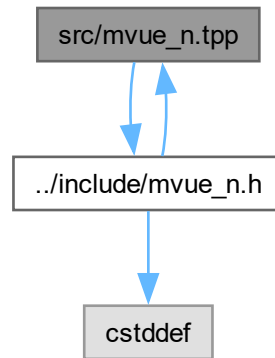
```

00001 #include "../include/mvue_4.h"
00002
00003 template <typename T>
00004 estimators::mvue::MVUE_4<T>::MVUE_4()
00005 {
00006 }
00007
00008 template <typename T>
00009 estimators::mvue::MVUE_4<T>::~~MVUE_4()
00010 {
00011 }
00012
00013 template <typename T>
00014 void estimators::mvue::MVUE_4<T>::init(T var_1_, T var_2_, T var_3_, T var_4_)
00015 {
00016     set_param(var_1_, var_2_, var_3_, var_4_);
00017 }
00018
00019 template <typename T>
00020 void estimators::mvue::MVUE_4<T>::set_param(T var_1_, T var_2_, T var_3_, T var_4_)
00021 {
00022     var_1 = var_1_;
00023     var_2 = var_2_;
00024     var_3 = var_3_;
00025     var_4 = var_4_;
00026 }
00027
00028 template <typename T>
00029 T estimators::mvue::MVUE_4<T>::update(T x1_i, T x2_i, T x3_i, T x4_i)
00030 {
00031     return ((x1_i / var_1) + (x2_i / var_2) + (x3_i / var_3) + (x4_i / var_4)) / ((1.0 / var_1) + (1.0
00032 / var_2) + (1.0 / var_3) + (1.0 / var_4));
00033 }
00034 template <typename T>
00035 void estimators::mvue::MVUE_4<T>::reset()
00036 {
00037 }
  
```

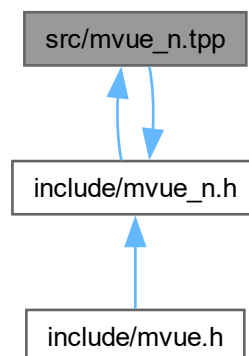
12.19 src/mvue_n.hpp File Reference

```
#include "../include/mvue_n.h"
```

Include dependency graph for mvue_n.hpp:



This graph shows which files directly or indirectly include this file:



12.20 mvue_n.hpp

[Go to the documentation of this file.](#)

```

00001 #include "../include/mvue_n.h"
00002
00003 template <typename T, std::size_t N>
00004 estimators::mvue::MVUE_N<T, N>::MVUE_N()
00005 {
00006 }
  
```

```
00007
00008 template <typename T, std::size_t N>
00009 estimators::mvue::MVUE_N<T, N>::~MVUE_N()
00010 {
00011 }
00012
00013 template <typename T, std::size_t N>
00014 void estimators::mvue::MVUE_N<T, N>::init(T *var_)
00015 {
00016     set_param(var_);
00017 }
00018
00019 template <typename T, std::size_t N>
00020 void estimators::mvue::MVUE_N<T, N>::set_param(T *var_)
00021 {
00022     var = var_;
00023 }
00024
00025 template <typename T, std::size_t N>
00026 T estimators::mvue::MVUE_N<T, N>::update(const T *x_i)
00027 {
00028     T nominator = 0.0;
00029     T denominator = 0.0;
00030
00031     for (int j = 0; j < N; j++)
00032     {
00033         nominator += x_i[j] / var[j];
00034         denominator += 1.0 / var[j];
00035     }
00036
00037     return nominator / denominator;
00038 }
00039
00040 template <typename T, std::size_t N>
00041 void estimators::mvue::MVUE_N<T, N>::reset()
00042 {
00043 }
```


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